



Federal Urdu University of Arts, Science and Technology
Islamabad

Multi-Vector OSINT Analyzer

PREPARED For ISeeWaves Pvt Ltd

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1.CHAPTER INCEPTION

PROPOSAL

FOR

Multi-Vector OSINT Analyzer

Version 1.0

Prepared by

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&

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21 Oct, 2025

Revision History

Version	Description	Author	Date
1.0	This document contains Project Proposal of Multi-Vector OSINT Analyzer	Mansoor khan & Atif Mehmood	21 th October, 2025

1. Introduction

In today's technology-driven era, the explosion of publicly available data has made Open Source Intelligence (OSINT) a cornerstone of modern investigation and cybersecurity analysis. By gathering data from websites, social media, and public databases, OSINT helps identify potential risks, data leaks, and identity exposures.

However, a significant gap exists in the current landscape: most OSINT tools are fragmented and designed for narrow, single-purpose tasks. This forces cybersecurity professionals to manually switch between multiple platforms, leading to an inefficient and time-consuming investigative process.

The Multi-Vector OSINT Analyzer, developed in collaboration with iSeeWaves Pvt Ltd, addresses this challenge by integrating diverse intelligence functions into a single, unified web platform. This system streamlines automated investigations across several key areas:

- Domain & Network: WHOIS Lookups and Subdomain Discovery.
- Threat Intelligence: Email and Password Breach Analysis, and URL Risk Detection.
- Identity & Media: Username OSINT and Image Analysis.

Furthermore, the platform includes a Password Encoder/Hasher (supporting MD5, SHA-1, and SHA-256) to promote secure data handling and password security awareness. By consolidating these capabilities, the Multi-Vector OSINT Analyzer provides a fast, ethical, and practical solution for digital forensics and investigative research, directly aligned with iSeeWaves' mission of advancing privacy-centric security technologies.

1.1 Problem statement

In the modern digital landscape, the massive growth of publicly available information has created a dual-edged sword: while valuable for threat intelligence, it introduces complex privacy and investigative challenges.

The primary issue is that current OSINT methodologies are highly fragmented. Professionals must rely on multiple independent tools for tasks like WHOIS lookups, subdomain discovery, and breach analysis. This independent operation forces users to:

- Switch between various platforms constantly.
- Manually correlate data from inconsistent outputs.

- Waste significant time linking findings, which increases the risk of inaccurate analysis.

Furthermore, most existing platforms lack integration and contextual understanding. They provide raw data without linking different intelligence sources, making it difficult for cybersecurity students and professionals to conduct seamless, end-to-end investigations.

To bridge this gap, there is a critical need for an integrated and automated OSINT system. Such a platform must consolidate diverse tools—handling domains, emails, images, and usernames—into a single, secure environment that prioritizes workflow efficiency and user privacy.

1.2 Proposed System

The Multi-Vector OSINT Analyzer, developed in collaboration with iSeeWaves Pvt Ltd, is a centralized web-based platform designed to unify fragmented intelligence operations. This system eliminates the need for multiple, disconnected tools by integrating major OSINT capabilities—such as WHOIS lookup, subdomain discovery, breach analysis, and URL risk detection—into a single dashboard. By consolidating these functions, the platform allows cybersecurity professionals and students to perform comprehensive investigations within a seamless, real-time environment.

Beyond core intelligence, the system includes a dedicated password encoding and hashing module supporting MD5, SHA-1, and SHA-256 algorithms. This feature not only enhances data security but also promotes cybersecurity awareness regarding safe password management. Overall, the platform provides a user-friendly interface where independent modules contribute to a structured analytical report. This unified approach significantly improves the efficiency, accuracy, and accessibility of digital investigations, directly supporting iSeeWaves' mission to deliver innovative and ethical security technologies.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
OSINT	Open Source Intelligence
WHOIS	Domain registration protocol
EXIF	Exchangeable Image File Format

Term	Definition
API	Application Programming Interface
SHA	Secure Hash Algorithm

1.4 Survey of the Related Applications

Before developing the Multi-Vector OSINT Analyzer, an in-depth survey of existing OSINT tools and platforms was carried out to analyze their scope, operational methods, and limitations. The main objective of this survey was to understand how current systems support open-source intelligence operations and to identify opportunities for improvement. The proposed project, developed for iSeeWaves Pvt Ltd, aims to address these limitations by combining multiple intelligence techniques into a single, user-friendly, and automated platform.

1.4.1 Maltego

Maltego, developed by Paterva, is one of the most recognized tools for OSINT and link analysis. It allows investigators to visualize complex relationships among people, domains, companies, and infrastructure through graph-based mapping. Its modular “transform” architecture makes it powerful for data correlation. However, Maltego’s advanced features come with a steep learning curve and depend heavily on manual configuration. The tool requires prior knowledge of investigation workflows and external data sources. Moreover, it lacks integrated features for password, email breach analysis, or URL risk assessment, limiting its use for general-purpose automation.

1.4.2 Shodan

Shodan is an internet search engine for connected devices that helps in identifying publicly accessible servers, IoT devices, and exposed infrastructure. It provides valuable insights into security risks by showing open ports, vulnerabilities. Although Shodan is useful for network reconnaissance, it does not perform identity-based analysis, image OSINT, or domain-specific intelligence in a consolidated manner. Its focus remains technical, making it less suitable for combined risk evaluations or for use by non-technical investigators.

1.4.3. Have I Been Pwned (HIBP)

Have I Been Pwned” is a widely used online platform that enables users to check whether their email addresses or passwords have appeared in publicly known data breaches. It utilizes verified breach datasets and is highly efficient for personal identity. Nevertheless, HIBP is a single-purpose platform, limited only to breach verification. It offers no functionality for cross-referencing with other OSINT data sources such as domains, usernames, or URLs. This limitation highlights the need for a broader, multi-vector analysis framework like the one proposed in this project.

1.4.4 SpiderFoot

SpiderFoot is an open-source automated OSINT tool capable of scanning hundreds of data sources. It supports modules for domains, IPs, and network infrastructure reconnaissance. Its modular system and automation potential make it powerful for data. However, SpiderFoot requires local setup, consumes significant resources, and lacks a centralized web dashboard for multi-user access. It also produces large volumes of unfiltered raw data, which can overwhelm non-expert users. A more streamlined and readable reporting system is essential for practical usability.

1.4.5 Recon-ng

Recon-ng is a modular framework used primarily for penetration testing and reconnaissance. It provides a command-line interface where modules can be executed to gather intelligence from public sources. While efficient for professionals, it lacks a graphical interface, user-friendly reporting, and modern visualization capabilities. This makes it unsuitable for educational, enterprise, or non-technical use cases.

1.4.6 OSINT Framework

The OSINT Framework is a web-based collection of categorized tools that guide users to relevant data sources. While it serves as an excellent starting point for investigations, it does not perform any active automation or analysis itself. Users must manually query each tool, collect the results, and compile them for further interpretation.

Comparison and Conclusion

From the above survey, it is evident that most existing OSINT tools specialize in specific domains of intelligence — such as image analysis, network reconnaissance, or breach verification — but lack integration and automation. Additionally, most platforms produce raw, technical outputs that require expert interpretation, limiting accessibility for non-technical users or small cybersecurity teams.

The Multi-Vector OSINT Analyzer, being developed for iSeeWaves Pvt Ltd, directly addresses these shortcomings by offering:

- A unified, web-based platform integrating multiple OSINT operations such as image intelligence, domain lookup, breach analysis, subdomain discovery, and URL risk assessment.
- A user-friendly dashboard with structured and readable results instead of raw data.
- An additional password encoding module that allows users to securely hash and encode passwords for testing or educational purposes.
- Strict compliance with ethical and passive OSINT practices, ensuring that all intelligence gathering is done through publicly available and legal sources.

Through this integrated approach, the system aims to provide faster investigations, simplified analysis, and improved accessibility, aligning with iSeeWaves Pvt Ltd's mission to develop ethical, intelligent, and privacy-preserving cybersecurity solutions.

2. Modules and Sub-Modules

The Multi-Vector OSINT Analyzer developed for iSeeWaves Pvt Ltd is designed as a modular and scalable web-based system, where each OSINT function is divided into individual components (modules). This structure enables easy maintenance, independent testing, and future integration of additional features or third-party APIs.

Each module performs a specific intelligence task, while the system as a whole provides a unified interface for managing and visualizing results efficiently.

2.1 Image OSINT Module

Purpose:

To analyze image files for hidden metadata and online presence.

Sub-Modules:

- **EXIF Data Extraction:** Extracts technical details like device model, camera settings, location (GPS), and timestamps from uploaded images.
- **Reverse Image Search:** Uses open APIs to check where similar or identical images appear on the internet, helping identify image sources or misuse.

2.2 WHOIS & Domain Intelligence Module

Purpose:

To gather detailed registration and ownership data of domains.

Sub-Modules:

- **WHOIS Lookup:** Fetches information such as registrar, creation/expiry dates, contact details, and DNS records.
- **DNS Record Viewer:** Displays domain A, MX, and NS records to help identify hosting or email servers.
- **Passive Footprinting:** Collects publicly available information without direct scanning or active probing.

2.3 Subdomain Finder Module

Purpose:

To identify all subdomains associated with a target domain using passive enumeration techniques.

Sub-Modules:

- **DNS Enumeration:** Collects subdomain entries from passive data sources.
- **Record Validation:** Verifies the existence of each subdomain to avoid duplicates or false results.

2.4 Email Breach Analysis Module

Purpose:

To verify whether an email address has appeared in publicly known data breaches.

Sub-Modules:

- **Breach Check API Integration:** Connects with breach databases to identify compromised credentials.
- **Breach Summary Report:** Displays the number, type, and year of breaches detected.

2.5 Password Breach Analysis Module

Purpose:

To test the exposure level of passwords using K-Anonymity APIs while maintaining privacy.

Sub-Modules:

- **Hashing & Prefix Search:** Encrypts passwords into SHA-1 hashes and checks partial matches against known leak databases.
- **Result Interpretation:** Returns the number of times a password hash was found in breaches.

2.6 Username OSINT Module

Purpose:

To search for a specific username across multiple social platforms and websites.

Sub-Modules:

- **Platform Scanner:** Checks for username existence across major platforms (GitHub, Twitter, Reddit, etc.).
- **Result Filter:** Classifies findings as "active," "inactive," or "not found" for clarity.

2.7 URL Risk Detection Module

Purpose:

To evaluate URLs for potential phishing, malware, or suspicious behavior.

Sub-Modules:

- **Lexical Analysis:** Analyzes the structure of URLs (length, special characters, domain age, etc.) to identify risky patterns.
- **Blacklist Verification:** Cross-checks URLs with trusted phishing and malware databases for threat validation.

2.8 Password Encoder & Hashing Module

Purpose:

To allow users to encode or hash passwords or strings securely for educational or verification purposes.

Sub-Modules:

- **Hash Generator:** Converts plain text into multiple hash formats (MD5, SHA-1, SHA-256).
- **Encoding Converter:** Supports Base64 and URL encoding for security training and testing.
- **Hash Verifier:** Optionally compares two hashes to confirm integrity or match.

This feature was added on request to enhance the system's educational value and to help iSeeWaves Pvt Ltd users demonstrate cryptographic transformations safely without real credential storage.

3.Primary Actors

Administrator

Security Analyst / Investigator

4. Tool and Technologies

The proposed Multi-Vector OSINT Analyzer is built primarily using open-source, command-line interface (CLI) tools and supporting technologies to ensure efficiency, flexibility, and transparency. These tools help perform real-time intelligence gathering and data analysis from publicly available sources.

4.1 Tools

The system integrates multiple open-source CLI tools, each responsible for specific intelligence tasks:

- **WHOIS CLI Tool** – Used to fetch domain ownership and registration information.
- **Sublist3r / Amass** – For subdomain enumeration and discovery.
- **theHarvester** – To gather email, username, and domain-related data from search engines and public sources.
- **ExifTool** – For image OSINT and metadata extraction.
- **Have I Been Pwned CLI / BreachCheck** – To check email and password breach data.
- **Hashlib (Python library)** – For password encoding and hashing using algorithms like MD5, SHA-1, and SHA-256.
- **Curl / Wget** – For data fetching and testing URL risk or phishing detection.

4.2 Development Tools

- **Python / Django** – Used for backend integration and automation of CLI tools.
- **HTML, CSS, JavaScript (Bootstrap)** – For front-end web interface.
- **SQLite / PostgreSQL** – For storing results and user history.
- **Git & GitHub** – For version control and collaborative development.
- **Linux Terminal / Kali Linux Environment** – For running and testing open-source OSINT tools.

4.3 Techniques

- **API Integration:** Some CLI tools are wrapped with RESTful APIs to interact with the web interface.
- **Automation Scripts:** Python scripts automate command execution and result parsing.

- **Data Visualization:** Results from multiple tools are organized into easy-to-understand tables and visual reports.
- **Security & Privacy:** Hashing and secure handling of user inputs ensure ethical and safe OSINT use.

5. Process Model and Design Approach

This project follows the Incremental Development Model, where each OSINT feature is developed and tested step by step. In this model, one module (like Image OSINT or Email Breach Analysis) is completed, tested, and then integrated with others. This helps in finding errors early and allows updates easily.

Process Model

- The project is divided into small parts (modules).
- Each module is developed, tested, and added to the main system.
- After all modules are ready, they are combined into one complete OSINT Analyzer.
- The model ensures smooth progress and reduces overall project risk.

Design Approach

- The system has three main layers:
 1. **Frontend:** Simple web interface for users to enter inputs (like domain, email, or image).
 2. **Backend:** Runs CLI-based open-source OSINT tools, collects and processes results.
 3. **Database:** Stores non-sensitive output data and logs safely.

All tools used in this project are open-source and perform passive information gathering only. The design is modular, secure, and easy to update with new OSINT tools in the future.

2.CHAPTER ELABORATION

Software Requirement Specifications

Multi-Vector OSINT Analyzer

Version 1.0

Prepared by

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11th Nov, 2025

Revision History

Version	Description	Author	Date
1.0	This covers the major SRS documents	Mansoor Khan / Atif Mehmood	11 th Nov, 2025

1. Introduction

1.1 Purpose

This SRS document describes the functional and non-functional requirements for the OSINT (Open-Source Intelligence) Platform. It is intended for developers, testers, security analysts, administrators, and project stakeholders.

1.2 Scope

The OSINT Platform is a web-based application for intelligence gathering and analysis. Key modules include:

- User authentication and management
- WHOIS lookup and subdomain enumeration
- Email/password breach checking
- Image metadata extraction
- Username search across platforms
- URL risk detection
- Password hashing/encoding
- Report generation
- Admin dashboard

1.3 Definitions and Acronyms

Term	Definition
OSINT	Open Source Intelligence
WHOIS	Domain registration protocol
EXIF	Exchangeable Image File Format
API	Application Programming Interface
SHA	Secure Hash Algorithm

2. Overall Description

2.1 Product Perspective

A standalone web-based system integrating multiple OSINT tools with external APIs for breach databases, WHOIS services, DNS servers, and social media platforms.

System Components:

- Frontend (User Interface)
- Backend (Application Logic)
- Database (User data & results)
- External APIs (Third-party services)

2.2 User Characteristics

User Type	Technical Expertise
Security Analyst	Intermediate to Advanced
Administrator	Advanced

3. Specific Requirements

3.1 Functional Requirements

3.1.1 User Registration

SRS ID	Requirement
SRS-1	System shall allow users to register with name, email, organization, and password
SRS-2	System shall validate mandatory fields and email format
SRS-3	System shall ensure unique email addresses

SRS ID	Requirement
SRS-4	System shall enforce password strength (8+ chars, 1 uppercase, 1 number, 1 special character)
SRS-5	System shall send verification email upon registration
SRS-6	System shall store credentials using bcrypt/Argon2 encryption

3.1.2 User Login

SRS ID	Requirement
SRS-7	System shall provide login with email/username and password
SRS-8	System shall validate credentials against database
SRS-9	System shall create secure session tokens upon authentication
SRS-10	System shall redirect to role-based dashboard after login
SRS-11	System shall prevent login for unverified accounts
SRS-12	System shall lock account after 5 failed login attempts for 15 minutes
SRS-13	System shall log all login attempts with timestamp and IP

3.1.3 WHOIS Lookup

SRS ID	Requirement
SRS-14	System shall validate domain format before WHOIS query
SRS-15	System shall display registrar, creation/expiry dates, name servers, contact info, and DNS records

SRS ID	Requirement
SRS-16	System shall handle unavailable WHOIS data with error messages
SRS-17	System shall save results to database and export in PDF/CSV/JSON

3.1.4 Subdomain Enumeration

SRS ID	Requirement
SRS-18	System shall use Sublist3r/Amass for subdomain discovery
SRS-19	System shall display progress indicator during scanning
SRS-20	System shall show subdomains with active/inactive status
SRS-21	System shall handle scan timeouts and empty results appropriately

3.1.5 Email Breach Check

SRS ID	Requirement
SRS-22	System shall integrate with Have I Been Pwned API
SRS-23	System shall display breach count, names, dates, and exposed data types
SRS-24	System shall handle API rate limits gracefully
SRS-25	System shall save and export breach reports

3.1.6 Password Breach Check

SRS ID	Requirement
SRS-26	System shall implement K-Anonymity (send only first 5 chars of SHA-1 hash)

SRS ID	Requirement
SRS-27	System shall use masked password input
SRS-28	System shall display breach frequency with severity warnings
SRS-29	System shall never store plain text passwords

3.1.7 Image Metadata Extraction

SRS ID	Requirement
SRS-30	System shall support JPG, PNG, TIFF, RAW formats (max 10MB)
SRS-31	System shall use ExifTool to extract camera model, GPS, date/time, software, resolution
SRS-32	System shall provide optional reverse image search

3.1.8 Username OSINT

SRS ID	Requirement
SRS-33	System shall use Sherlock/theHarvester for username search
SRS-34	System shall scan Twitter, GitHub, Reddit, Instagram, LinkedIn, Facebook, YouTube
SRS-35	System shall display status (Active/Inactive/Not Found) with profile links

3.1.9 URL Risk Detection

SRS ID	Requirement
SRS-36	System shall perform lexical analysis (length, special chars, domain age, HTTPS)
SRS-37	System shall query blacklist databases
SRS-38	System shall display risk level (Safe/Suspicious/Dangerous) with justification

3.1.10 Password Hashing/Encoding

SRS ID	Requirement
SRS-39	System shall support MD5, SHA-1, SHA-256, SHA-512, Base64
SRS-40	System shall provide hash comparison functionality
SRS-41	System shall not store plain text for hashing operations

3.1.11 Generate & Export Report

SRS ID	Requirement
SRS-42	System shall auto-save all investigation results
SRS-43	System shall generate reports with summary, timestamps, module results, and recommendations
SRS-44	System shall export in PDF, CSV, JSON formats
SRS-45	System shall allow custom report module selection

3.1.12 Admin - Manage Users

SRS ID	Requirement
SRS-46	System shall display all users with status (Active/Pending/Suspended)
SRS-47	System shall allow approve/suspend/delete/change role actions
SRS-48	System shall send email notifications on status changes
SRS-49	System shall prevent self-deletion and require confirmation for admin privileges
SRS-50	System shall log all admin actions

3.1.13 Admin - Monitor System Logs

SRS ID	Requirement
SRS-51	System shall log all activities with timestamp, user, action, module, IP, status
SRS-52	System shall filter logs by date, user, module, action type
SRS-53	System shall flag suspicious activities (failed logins, unusual patterns)
SRS-54	System shall export logs in CSV/PDF and implement 90-day retention policy

3.2 Non-Functional Requirements

3.2.1 Performance

The system is designed to deliver high efficiency and responsiveness to ensure a smooth user experience. The dashboard is required to load within 3 seconds, providing immediate access to the core features. To handle academic and professional traffic, the system will support up to 100 concurrent users without degradation in service. For core intelligence operations, specifically WHOIS lookups, 95% of requests must be completed within a 5-second window to maintain high-speed investigative workflows.

3.2.2 Security

Security is a critical pillar of the Multi-Vector OSINT Analyzer. All data transmissions will be secured using HTTPS/TLS encryption to prevent eavesdropping. To protect user sessions, the system will implement an automatic session timeout after 30 minutes of inactivity. Robust protection mechanisms will be integrated to safeguard the platform against common web vulnerabilities, including SQL injection, CSRF, and XSS attacks. Additionally, rate limiting of 100 requests per hour for resource-heavy operations will be enforced to prevent abuse, while ensuring full compliance with GDPR and CCPA data privacy standards.

3.2.3 Reliability

The reliability of the platform ensures that investigative tools are available whenever needed. The system targets a minimum uptime of 99.5%, backed by daily database backups to prevent data loss. In the event of a system failure, recovery protocols are designed to restore full functionality within 5 minutes, ensuring minimal disruption to the user's research or investigation.

3.2.4 Usability

The platform prioritizes user experience through an intuitive UI that requires minimal training for new users. The interface is fully responsive, ensuring seamless functionality across desktops, tablets, and mobile devices. To assist users during complex operations, the system will provide clear, descriptive error messages and consistent visual feedback for every action performed within the dashboard.

3.2.5 Compatibility

To ensure wide accessibility, the system is engineered to support all modern web browsers, including Chrome 90+, Firefox 88+, Safari 14+, and Edge 90+. Furthermore, the web-based application is designed to function across multiple operating systems, including Windows 10+, macOS 10.15+, and various Linux distributions, providing a consistent experience regardless of the user's hardware environment.

Use Case Diagram

Multi-Vector OSINT Analyzer

Version 1.0

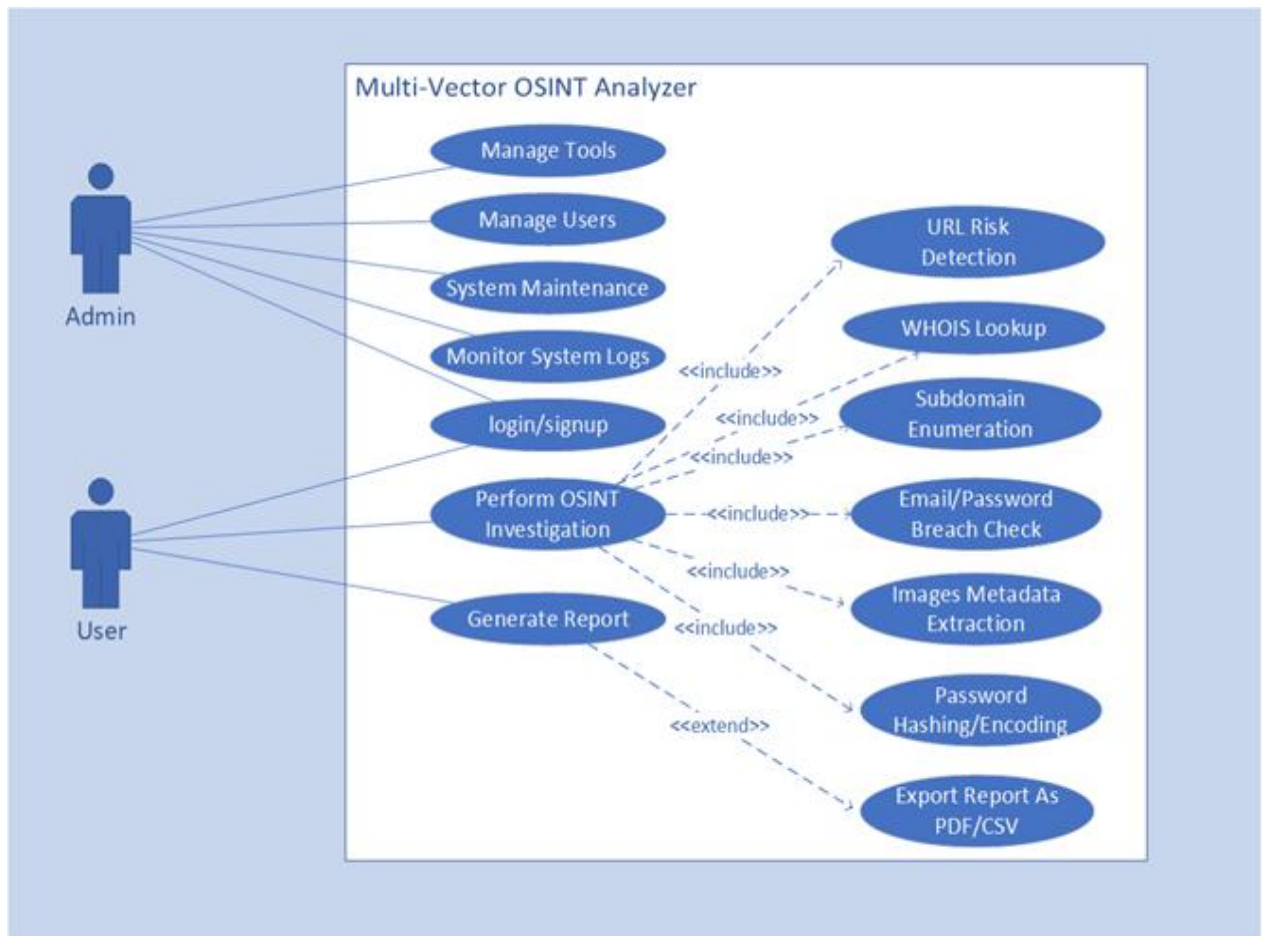
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Fully Dressed Use Cases

Multi-Vector OSINT Analyzer

Version 1.0

Prepared by

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23th Nov, 2026

Revision History

Version	Description	Author	Date
1.0	This cover the User cases of the system	Mansoor Khan Atif Mehmood	23 st Nov, 2026

UC-01 User Registration

Field	Details
Use Case ID	01
Use Case Name	User Registration
Use Case Description	This use case describes how a new Security Analyst or user registers for an account to access the OSINT platform.
Prepared by	Mansoor Khan & Atif Mehmood
Date	27th November, 2025
Primary Actors	User
Stakeholders & Interests	User: Wants to create an account to start using OSINT tools. System Admin: Wants to ensure only valid users register with secure passwords.
Pre-Conditions	• System must be online. • User must have a valid email address.

Main Success Scenario

User Action	System Response
1. User navigates to the Registration page.	System displays registration form.
2. User enters details (Name, Email, Organization, Password).	System validates email format and password strength (8+ chars, 1 uppercase, 1 special).
3. User submits the registration form.	System stores user data securely (hashed password) and sends a confirmation email.
4. User checks email for verification link.	System activates account upon verification.

Extension Points / Alternative Way

2a. If email is already registered:

- System prompts: "Email already exists. Please login."

2b. If password is weak:

- System prompts: "Password must contain 8+ characters, uppercase, and symbol."

Post-Conditions

- User account is created and stored in the database.
- User is ready to log in.

UC-02 User Login

Field	Details
Use Case ID	02
Use Case Name	User Login
Use Case Description	This use case describes how a registered user (Security Analyst or Admin) logs into the OSINT platform to access available tools and features.
Prepared by	Mansoor Khan & Atif Mehmood
Date	3rd January, 2026
Primary Actors	User, Admin
Stakeholders & Interests	User: Wants secure and quick access to OSINT tools. Admin: Wants to monitor user activity and ensure authorized access only.
Pre-Conditions	• User must have a registered and verified account. • System must be online and accessible.

Main Success Scenario

User Action	System Response
1. User navigates to the Login page.	System displays login form (Email/Username & Password fields).

User Action	System Response
2. User enters valid credentials and clicks "Login".	System validates credentials against database.
3. User is redirected to dashboard.	System authenticates user, creates session token, and displays user dashboard with available OSINT tools.

Extension Points / Alternative Way

2a. If credentials are incorrect:

- System prompts: "Invalid email or password. Please try again."

2b. If account is not verified:

- System prompts: "Please verify your email before logging in."

Post-Conditions

- User is successfully logged in with an active session.
- User can access OSINT modules based on their role.

UC-03 Perform WHOIS Lookup

Field	Details
Use Case ID	03
Use Case Name	Perform WHOIS Lookup
Use Case Description	This use case describes how a user performs a WHOIS lookup to gather domain registration and ownership information.
Prepared by	Mansoor Khan & Atif Mehmood
Date	3rd January, 2026
Primary Actors	Security Analyst, Admin

Field	Details
Stakeholders & Interests	Security Analyst: Wants to gather intelligence on domain ownership, registration dates, and DNS records. Admin: Wants to ensure tool reliability and log usage.
Pre-Conditions	• User must be logged in.
 • System must have access to WHOIS CLI tool and internet connectivity.

Main Success Scenario

User Action	System Response
1. User selects "WHOIS Lookup" from dashboard.	System displays input form for domain name.
2. User enters target domain (e.g., example.com).	System validates domain format.
3. User clicks "Lookup".	System executes WHOIS CLI command and fetches data.

Extension Points / Alternative Way

2a. If domain format is invalid:

- System prompts: "Please enter a valid domain name (e.g., example.com)."

3a. If domain does not exist or WHOIS data unavailable:

- System displays: "No WHOIS information found for this domain."

Post-Conditions

- WHOIS data is retrieved and displayed.
- Results are optionally saved to database or exported.

UC-04 Subdomain Enumeration

Field	Details
Use Case ID	04
Use Case Name	Subdomain Enumeration
Use Case Description	This use case describes how a user discovers all subdomains associated with a target domain using passive enumeration techniques.
Prepared by	Mansoor Khan & Atif Mehmood
Date	3rd January, 2026
Primary Actors	Security Analyst, Admin
Stakeholders & Interests	Security Analyst: Wants to identify all subdomains for reconnaissance and vulnerability assessment. Admin: Wants to track tool usage and performance.
Pre-Conditions	• User must be logged in.
 • System must have Sublist3r/Amass tool configured.
 • Internet connection must be available.

Main Success Scenario

User Action	System Response
1. User selects "Subdomain Finder" from dashboard.	System displays input form for target domain.
2. User enters domain name (e.g., example.com).	System validates input format.
3. User clicks "Find Subdomains".	System runs subdomain enumeration using CLI tool (Sublist3r/Amass).
4. User waits for scanning process.	System performs DNS enumeration, validates subdomains, and displays progress indicator.

User Action	System Response
5. User views discovered subdomains list.	System displays list of subdomains with status (active/inactive).
6. User saves or exports results.	System stores results in database and/or exports as requested format.

Extension Points / Alternative Way

2a. If domain format is invalid:

- System prompts: "Invalid domain format. Please enter a valid domain."

4a. If no subdomains are found:

- System displays: "No subdomains discovered for this domain."

4b. If enumeration process times out:

- System displays: "Scan timeout. Please try again or use a smaller scope."

Post-Conditions

- List of discovered subdomains is displayed and saved.
- User can perform further analysis on specific subdomains.

UC-05 Email Breach Check

Field	Details
Use Case ID	05
Use Case Name	Email Breach Check
Use Case Description	This use case describes how a user checks if an email address has been compromised in known data breaches.
Prepared by	Mansoor Khan & Atif Mehmood

Field	Details
Date	3rd January, 2026
Primary Actors	Security Analyst, User
Stakeholders & Interests	User: Wants to verify if their email or target email has been exposed in breaches. Security Analyst: Wants to assess risk level of compromised credentials.
Pre-Conditions	• User must be logged in.
 • System must have API access to breach databases (Have I Been Pwned or similar).
 • Internet connection required.

Main Success Scenario

User Action	System Response
1. User selects "Email Breach Check" from dashboard.	System displays input form for email address.
2. User enters target email address.	System validates email format.
3. User clicks "Check Breach".	System queries breach database API using entered email.
4. User views breach report.	System displays: Number of breaches found, Breach names, Dates of breaches, Types of data exposed.
5. User saves or exports report.	System saves results to database and/or exports as PDF/CSV.

Extension Points / Alternative Way

2a. If email format is invalid:

- System prompts: "Please enter a valid email address."

4a. If no breaches found:

- System displays: "Good news! This email has not been found in any known breaches."

3a. If API rate limit exceeded:

- System prompts: "Too many requests. Please wait a few minutes and try again."

Post-Conditions

- Breach status of email is displayed with detailed report.
- Results are saved for future reference.

UC-06 Password Breach Check

Field	Details
Use Case ID	06
Use Case Name	Password Breach Check
Use Case Description	This use case describes how a user checks if a password has been exposed in data breaches using K-Anonymity method (SHA-1 hash prefix matching).
Prepared by	Mansoor Khan & Atif Mehmood
Date	3rd January, 2026
Primary Actors	Security Analyst, User
Stakeholders & Interests	User: Wants to verify password strength and exposure risk without revealing actual password. Security Analyst: Wants to educate users about password security.
Pre-Conditions	• User must be logged in.
 • System must have access to password breach API (Pwned Passwords).
 • Hashing library (SHA-1) must be available.

Main Success Scenario

User Action	System Response
1. User selects "Password Breach Check" from dashboard.	System displays secure input field for password.

User Action	System Response
2. User enters password to check.	System does not display password (masked input).
3. User clicks "Check Password".	System hashes password using SHA-1, sends first 5 characters of hash to API (K-Anonymity).
4. User views exposure report.	System displays: "This password has been seen X times in data breaches" or "Password not found in breaches".
5. User acknowledges result.	System clears password from memory and offers password security tips.

Extension Points / Alternative Way

2a. If password field is empty:

- System prompts: "Please enter a password to check."

4a. If API connection fails:

- System prompts: "Unable to connect to breach database. Please try again later."

4b. If password found multiple times:

- System warns: "CRITICAL: This password is highly compromised. Change it immediately!"

Post-Conditions

- Password exposure level is displayed without storing actual password.
- User is aware of password risk level.

Fully Dressed Use Case: UC-07 Image Metadata Extraction

Field	Details
Use Case ID	07
Use Case Name	Image Metadata Extraction (EXIF)
Use Case Description	This use case describes how a user uploads an image to extract hidden metadata such as GPS location, camera details, and timestamps.
Prepared by	Mansoor Khan & Atif Mehmood
Date	3rd January, 2026
Primary Actors	Security Analyst, Investigator
Stakeholders & Interests	Investigator: Wants to gather forensic evidence from images. Security Analyst: Wants to identify image origins and manipulation.
Pre-Conditions	• User must be logged in.
 • System must have ExifTool installed.
 • Supported image formats: JPG, PNG, TIFF, RAW.

Main Success Scenario

User Action	System Response
1. User selects "Image OSINT" from dashboard.	System displays image upload interface.
2. User uploads image file.	System validates file format and size (max 10MB).
3. User clicks "Extract Metadata".	System runs ExifTool to extract EXIF data.
4. User views metadata report.	System displays: Camera Model, GPS Coordinates, Date/Time Taken, Software Used, Image Resolution, and other technical details.

User Action	System Response
5. User performs reverse image search (optional).	System uses Google/Bing reverse search API to find similar images online.
6. User saves or exports results.	System stores metadata report and exports as needed.

Extension Points / Alternative Way

2a. If file format is unsupported:

- System prompts: "Unsupported file format. Please upload JPG, PNG, TIFF, or RAW files."

2b. If file size exceeds limit:

- System prompts: "File too large. Maximum size is 10MB."

4a. If no EXIF data found:

- System displays: "No metadata found. Image may have been stripped or edited."

Post-Conditions

- Image metadata is extracted and displayed.
- Optional reverse search results are available.

UC-08 Username OSINT

Field	Details
Use Case ID	08
Use Case Name	Username Search Across Platforms
Use Case Description	This use case describes how a user searches for a specific username across multiple social media platforms and websites to gather digital footprint.
Prepared by	Mansoor Khan & Atif Mehmood

Field	Details
Date	3rd January, 2026
Primary Actors	Security Analyst, Investigator
Stakeholders & Interests	Investigator: Wants to trace online presence of a person or entity. Security Analyst: Wants to assess social engineering risks.
Pre-Conditions	• User must be logged in.
 • System must have access to username search tool (e.g., Sherlock, theHarvester).
 • Internet connection required.

Main Success Scenario

User Action	System Response
1. User selects "Username OSINT" from dashboard.	System displays input field for username.
2. User enters target username.	System validates input.
3. User clicks "Search Username".	System scans multiple platforms (Twitter, GitHub, Reddit, Instagram, LinkedIn, etc.) for username existence.
4. User views scan results.	System display00000000000000s list of platforms where username is found with status (Active/Inactive/Not Found) and profile links.
5. User saves or exports results.	System stores findings in database and exports report.

Extension Points / Alternative Way

2a. If username field is empty:

- System prompts: "Please enter a username to search."

4a. If username not found on any platform:

- System displays: "No matches found for this username across checked platforms."

3a. If scanning takes too long:

- System shows progress bar and platform-by-platform results as they arrive.

Post-Conditions

- List of platforms with username presence is displayed.
- User has links to active profiles for further investigation.

UC-09 URL Risk Detection

Field	Details
Use Case ID	09
Use Case Name	URL Risk Detection
Use Case Description	This use case describes how a user analyzes a URL to detect potential phishing, malware, or suspicious behavior.
Prepared by	Mansoor Khan & Atif Mehmood
Date	3rd January, 2026
Primary Actors	Security Analyst, User
Stakeholders & Interests	User: Wants to verify if a URL is safe before clicking. Security Analyst: Wants to identify malicious URLs for threat intelligence.
Pre-Conditions	• User must be logged in. • System must have access to URL analysis tools and blacklist databases. • Internet connection required.

Main Success Scenario

User Action	System Response
1. User selects "URL Risk Detection" from dashboard.	System displays input field for URL.

User Action	System Response
2. User enters suspicious URL.	System validates URL format.
3. User clicks "Analyze URL".	System performs lexical analysis (checks URL length, special characters, domain age, HTTPS status).
4. System checks against blacklist databases.	System queries known phishing/malware databases for URL reputation.
5. User views risk assessment report.	System displays: Risk Level (Safe/Suspicious/Dangerous), Reasons for classification, Domain age, SSL certificate status, Blacklist status.
6. User saves or exports report.	System logs analysis and exports report if needed.

Extension Points / Alternative Way

2a. If URL format is invalid:

- System prompts: "Please enter a valid URL (e.g., https://example.com)."

5a. If URL is flagged as dangerous:

- System displays warning: "⚠ DANGER: This URL is flagged for phishing/malware. Do not visit!"

4a. If blacklist database is unavailable:

- System performs only lexical analysis and notifies user: "Blacklist check unavailable. Results based on structural analysis only."

Post-Conditions

- URL risk level is determined and displayed with justification.
- User is informed about safety of the URL.

UC-10 Password Hashing/Encoding

Field	Details
Use Case ID	10
Use Case Name	Password Hashing and Encoding
Use Case Description	This use case describes how a user generates secure hashes or encodes text using various algorithms for educational or testing purposes.
Prepared by	Mansoor Khan & Atif Mehmood
Date	3rd January, 2026
Primary Actors	Security Analyst, User, Student
Stakeholders & Interests	User: Wants to understand password hashing for security awareness. Student: Wants to learn about cryptographic algorithms. Security Analyst: Wants to test hash integrity.
Pre-Conditions	• User must be logged in.
 • System must have hashing library (Hashlib in Python) available.

Main Success Scenario

User Action	System Response
1. User selects "Password Encoder/Hasher" from dashboard.	System displays input field for plain text/password and algorithm selection dropdown.
2. User enters text and selects algorithm (MD5/SHA-1/SHA-256/Base64).	System validates input.
3. User clicks "Generate Hash/Encode".	System processes input using selected algorithm.
4. User views generated hash/encoded string.	System displays result in text box with copy button.
5. User optionally compares two hashes.	System provides hash comparison tool to verify integrity or match.

User Action	System Response
6. User copies or saves result.	System allows copying to clipboard; does not store plain text passwords.

Extension Points / Alternative Way

2a. If input field is empty:

- System prompts: "Please enter text to hash or encode."

5a. If user enters two hashes for comparison:

- System compares and displays: "Hashes match ✓" or "Hashes do not match X"

Note: System never stores plain text passwords for security reasons.

Post-Conditions

- Hashed or encoded result is generated and displayed.
- User understands cryptographic transformation process.

UC-11 Generate & Export Report

Field	Details
Use Case ID	11
Use Case Name	Generate and Export OSINT Report
Use Case Description	This use case describes how a user generates a comprehensive OSINT investigation report and exports it in various formats.
Prepared by	Mansoor Khan & Atif Mehmood
Date	3rd January, 2026
Primary Actors	Security Analyst, Admin

Field	Details
Stakeholders & Interests	Security Analyst: Wants to compile all findings into a professional report. Admin: Wants to maintain investigation records for compliance.
Pre-Conditions	• User must have completed at least one OSINT analysis.
 • Results must be saved in database.

Main Success Scenario

User Action	System Response
1. User completes one or more OSINT investigations.	System automatically saves results to database.
2. User clicks "Generate Report" button.	System compiles all saved results into structured report format.
3. User selects report format (PDF/CSV/JSON).	System displays format selection dialog.
4. User confirms export.	System generates report file with: Investigation summary, Timestamp, Module-wise results, Risk assessments, Recommendations.
5. User downloads report file.	System provides download link and stores report in user's history.

Extension Points / Alternative Way

2a. If no results are available:

- System prompts: "No investigation data found. Please perform at least one OSINT analysis first."

4a. If user wants custom report:

- System allows user to select specific modules to include in report.

5a. If export fails due to system error:

- System prompts: "Export failed. Please try again or contact support."

Post-Conditions

- Complete OSINT report is generated and downloaded.
- Report is accessible in user's investigation history.

UC-12 Admin - Manage Users

Field	Details
Use Case ID	12
Use Case Name	Manage Users (Admin)
Use Case Description	This use case describes how an administrator manages user accounts, roles, and permissions on the OSINT platform.
Prepared by	Mansoor Khan & Atif Mehmood
Date	3rd January, 2026
Primary Actors	Administrator
Stakeholders & Interests	Administrator: Wants to control user access and maintain system security. Users: Want timely account approval and support.
Pre-Conditions	• Admin must be logged in with administrative privileges. • System must have user management module active.

Main Success Scenario

User Action	System Response
1. Admin navigates to "User Management" section.	System displays list of all registered users with status (Active/Pending/Suspended).
2. Admin selects a user to manage.	System displays user details: Name, Email, Organization, Registration Date, Activity Log.

User Action	System Response
3. Admin performs action: Approve/Suspend/Delete/Change Role.	System updates user status in database.
4. Admin saves changes.	System sends notification email to affected user and logs admin action.
5. Admin views updated user list.	System refreshes user management dashboard with updated information.

Extension Points / Alternative Way

3a. If admin tries to delete their own account:

- System prompts: "You cannot delete your own admin account. Please contact another administrator."

3b. If admin suspends a user:

- System immediately revokes user's session and blocks login access.

3c. If admin changes user role to Admin:

- System requires confirmation: "Are you sure you want to grant admin privileges to this user?"

Post-Conditions

- User account status or role is updated.
- User is notified of changes via email.
- Admin action is logged for audit purposes.

UC-13 Admin - Monitor System Logs

Field	Details
Use Case ID	13

Field	Details
Use Case Name	Monitor System Logs (Admin)
Use Case Description	This use case describes how an administrator views and monitors system activity logs, user actions, and security events.
Prepared by	Mansoor Khan & Atif Mehmood
Date	3rd January, 2026
Primary Actors	Administrator
Stakeholders & Interests	Administrator: Wants to track system usage, detect anomalies, and ensure compliance. Organization: Wants audit trails for security and legal purposes.
Pre-Conditions	• Admin must be logged in with administrative privileges.
 • System logging must be enabled and operational.

Main Success Scenario

User Action	System Response
1. Admin navigates to "System Logs" section.	System displays log dashboard with filters (Date Range, User, Module, Action Type).
2. Admin applies filters to view specific logs.	System queries database and displays filtered log entries.
3. Admin views log details.	System shows: Timestamp, User ID, Action Performed, Module Used, IP Address, Status (Success/Failure).
4. Admin exports logs for audit.	System generates log report in CSV/PDF format.
5. Admin reviews security alerts (if any).	System highlights suspicious activities (multiple failed logins, unusual access patterns).

Extension Points / Alternative Way

5a. If suspicious activity detected:

- System automatically flags entry and notifies admin via email/dashboard alert.

2a. If no logs match filter criteria:

- System displays: "No logs found for selected criteria."

4a. If admin wants to delete old logs:

- System requires confirmation and implements data retention policy (e.g., keep logs for 90 days).

Post-Conditions

- Admin has reviewed system activity.
- Security anomalies are identified and addressed.
- Logs are exported for compliance or audit purposes.

System Sequence Diagram

FOR

Multi-Vector OSINT Analyzer

Version 1.0

Prepared by

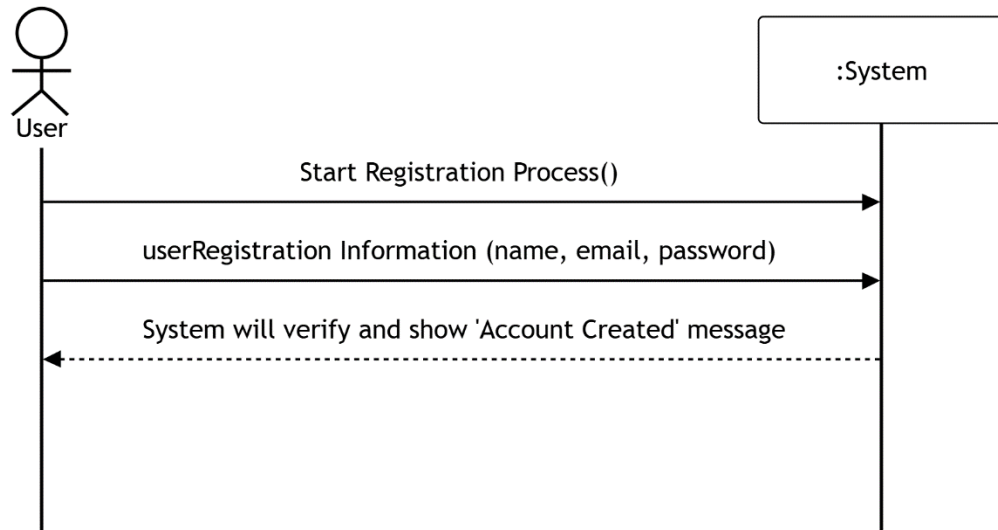
Mansoor khan

&

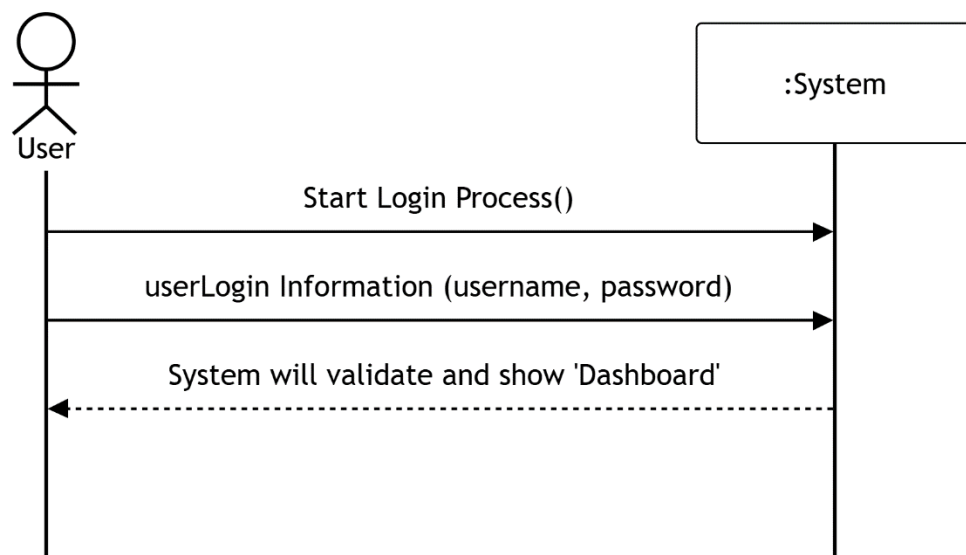
Atif Mehmood

01st Jan, 2026

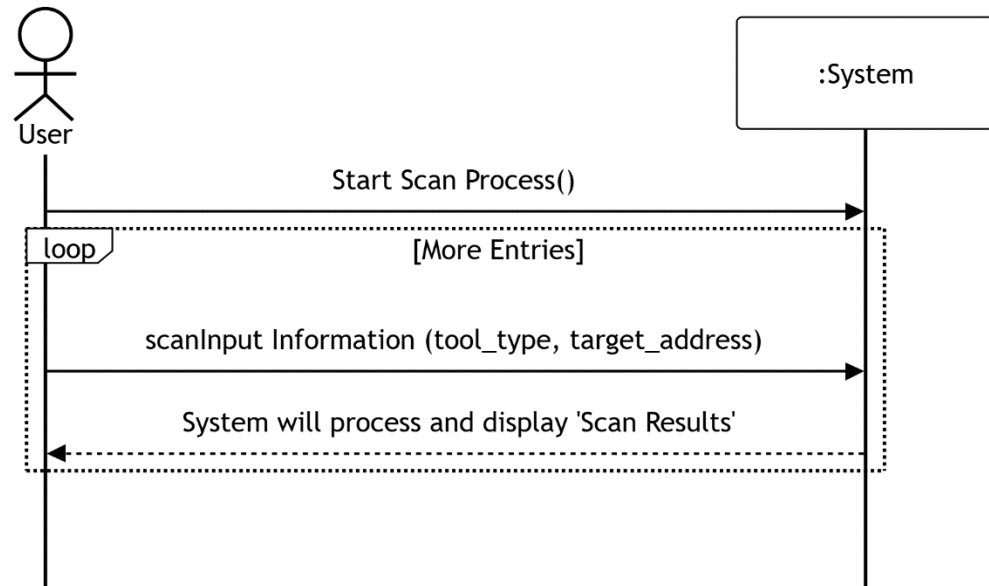
SSD 01: Process Registration



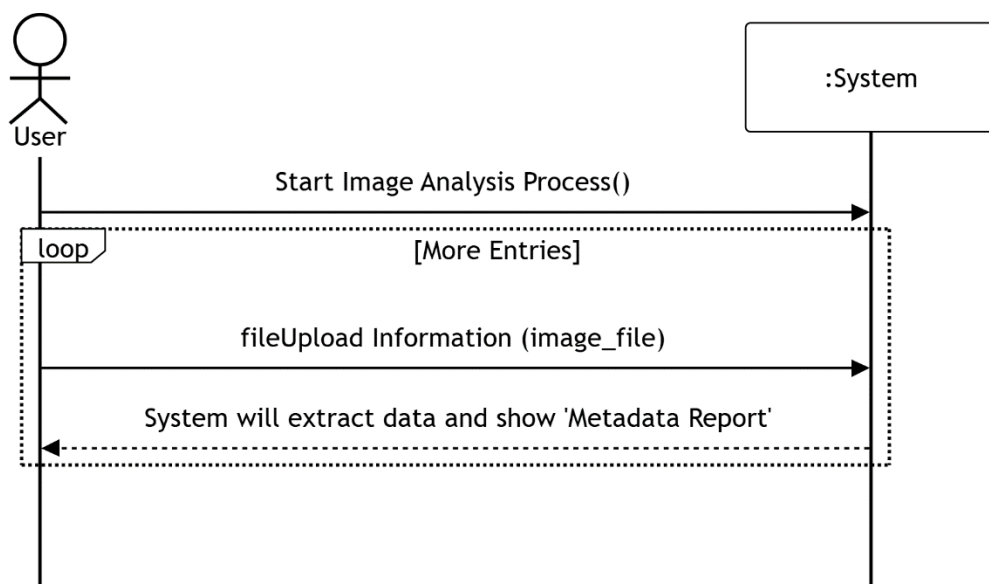
SSD 02: Process Login



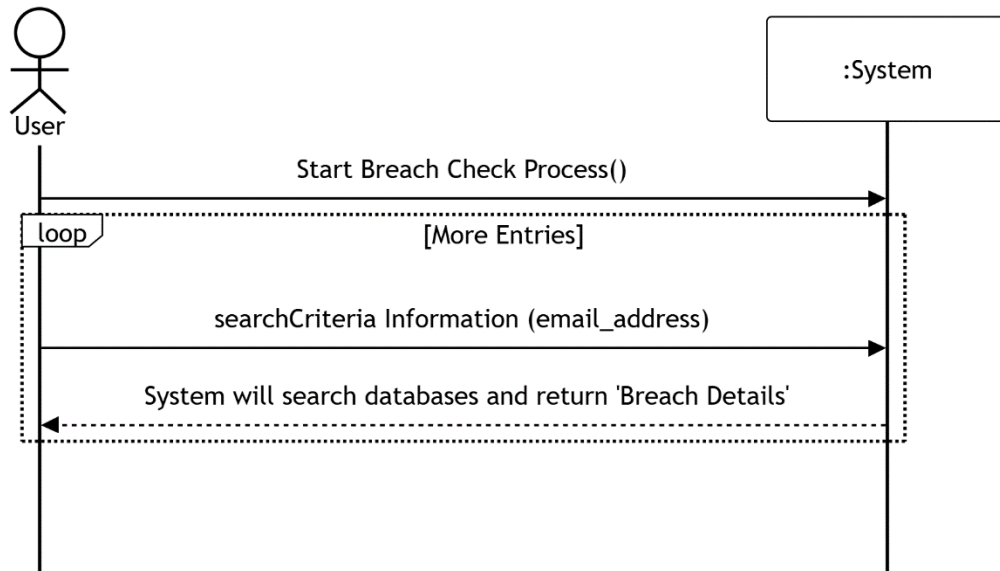
SSD 03: Perform OSINT Scan



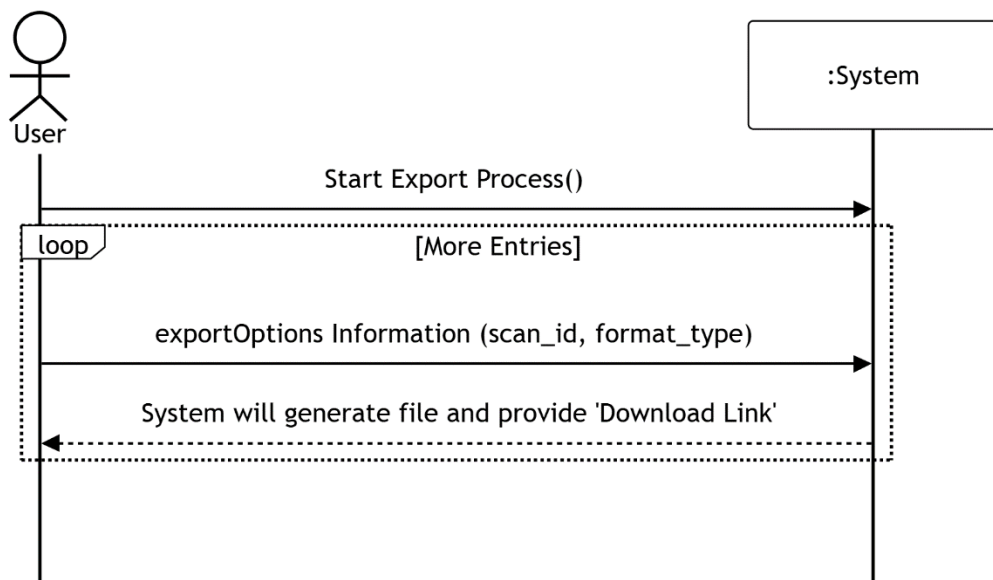
SSD 04: Extract Image Metadata



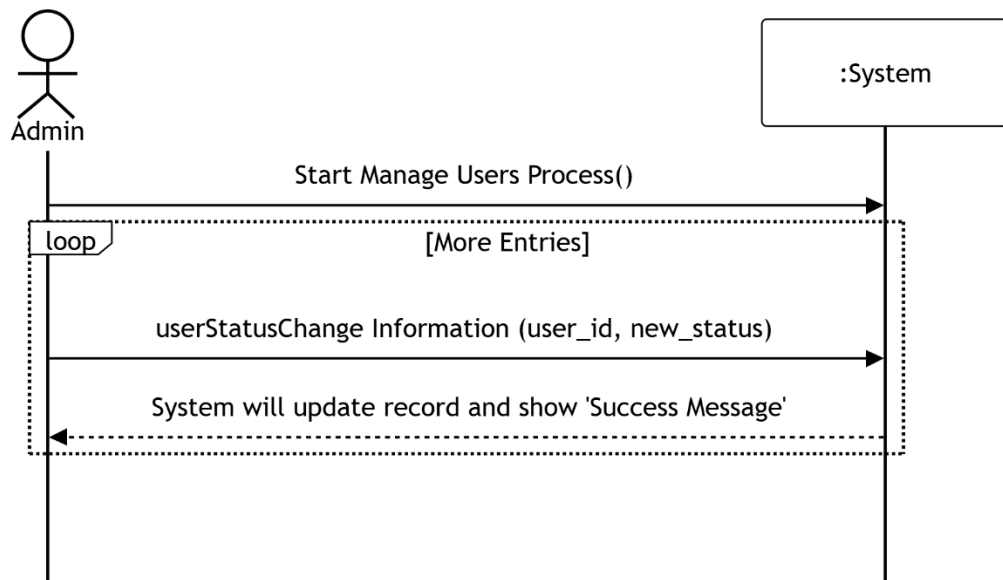
SSD 05: Check Email Breach



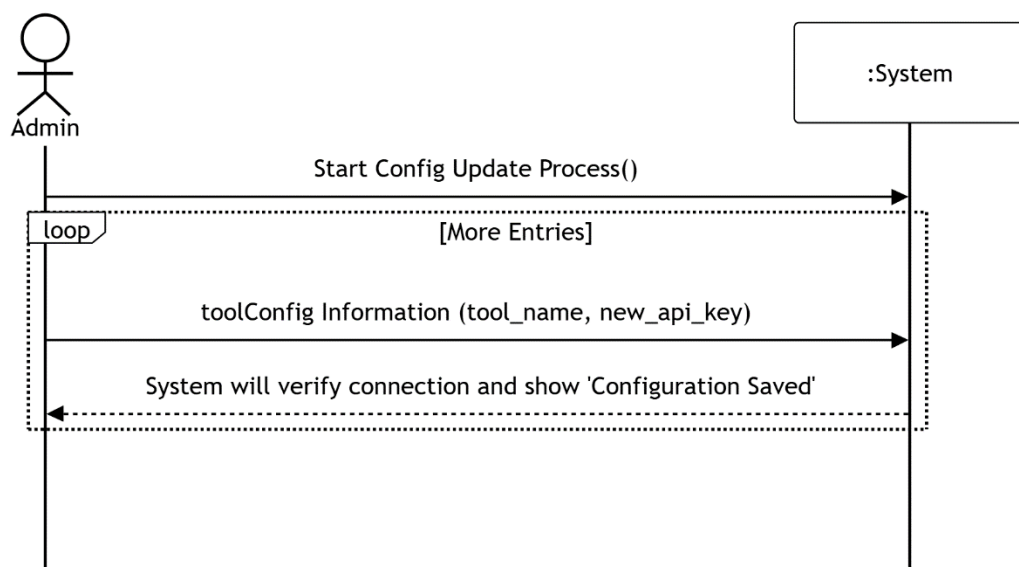
SSD 06: Generate Report



SSD 07: Manage Users



SSD 08: Manage Tools



Sequence Diagram

FOR

Multi-Vector OSINT Analyzer

Version 1.0

Prepared by

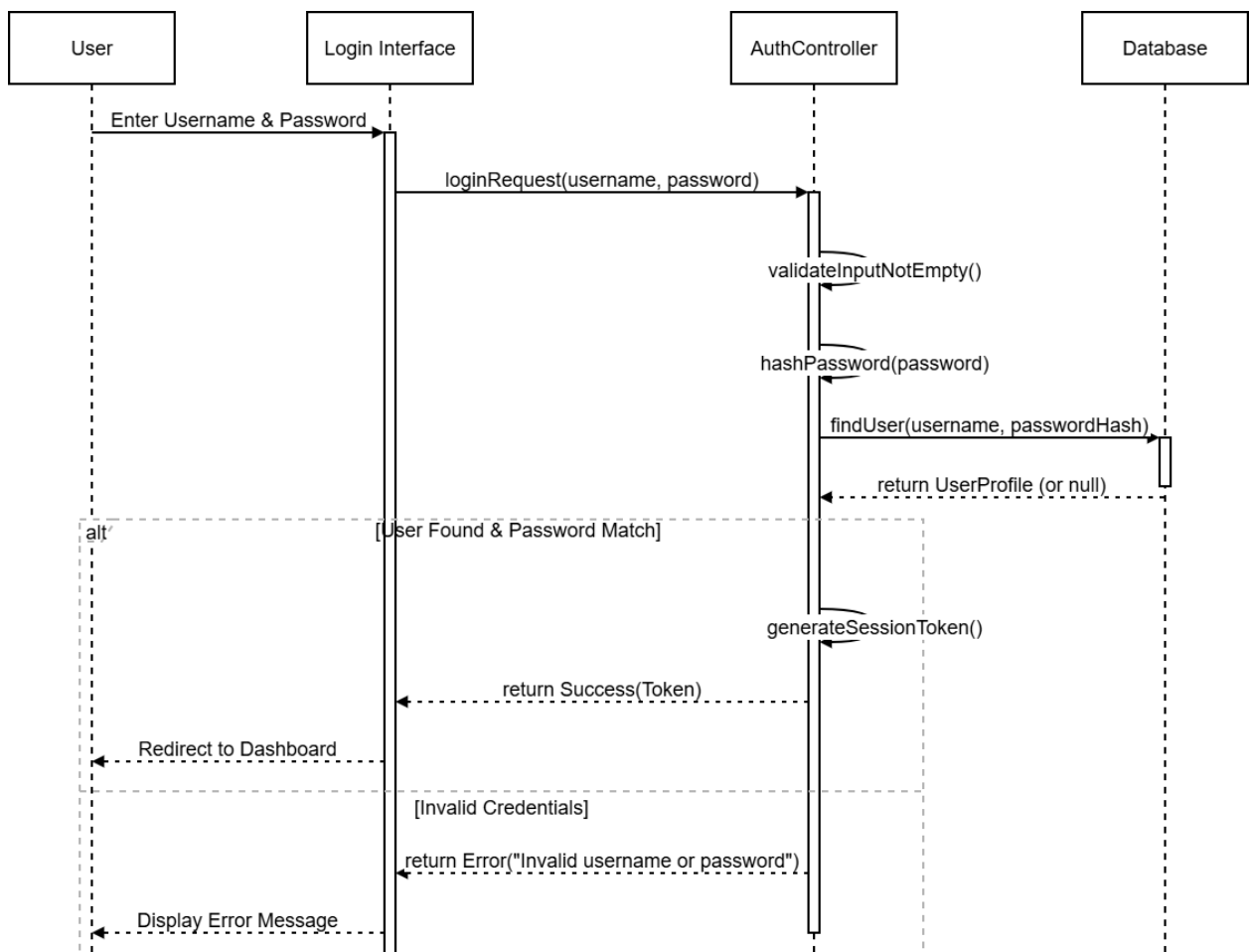
Mansoor Khan

&

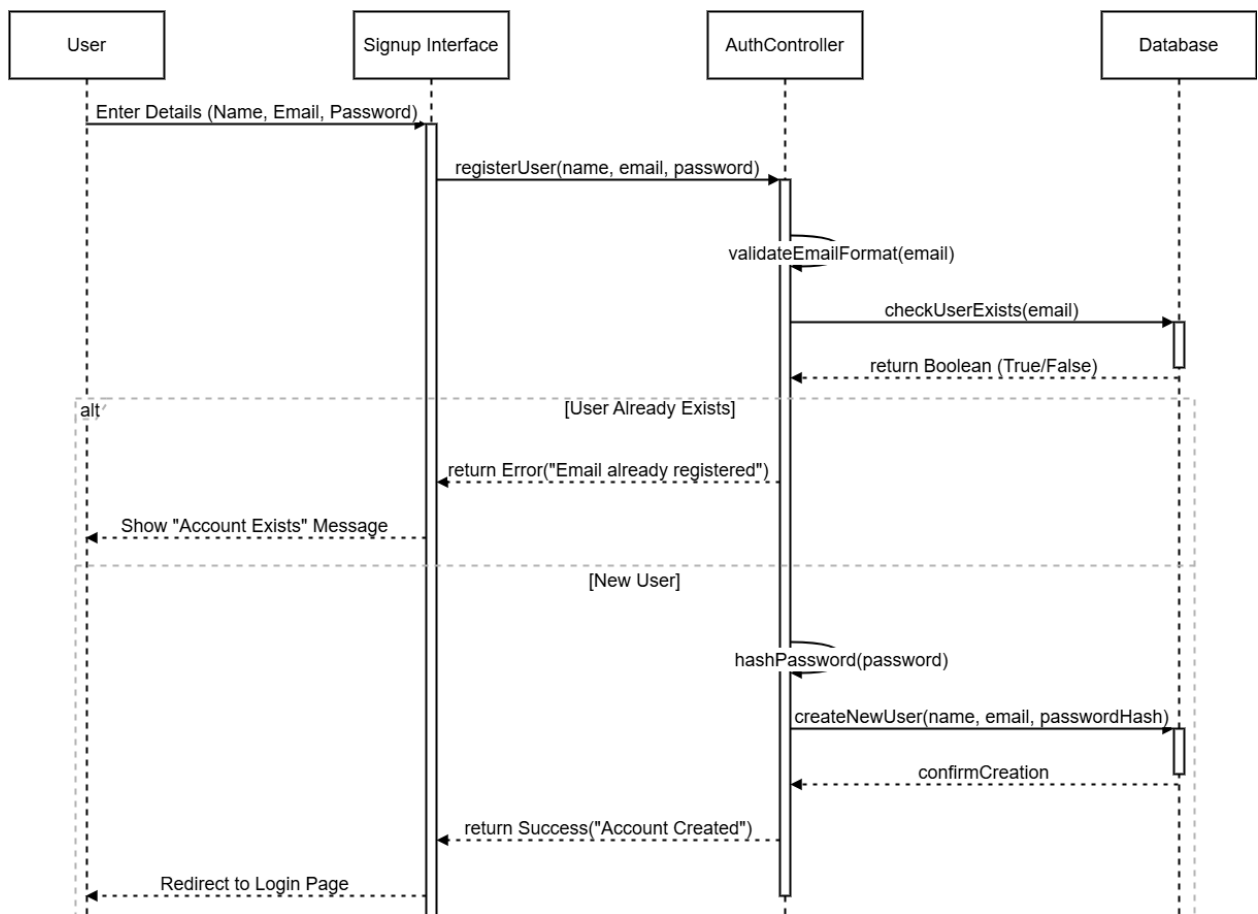
Atif Mehmood

7th Jan, 2026

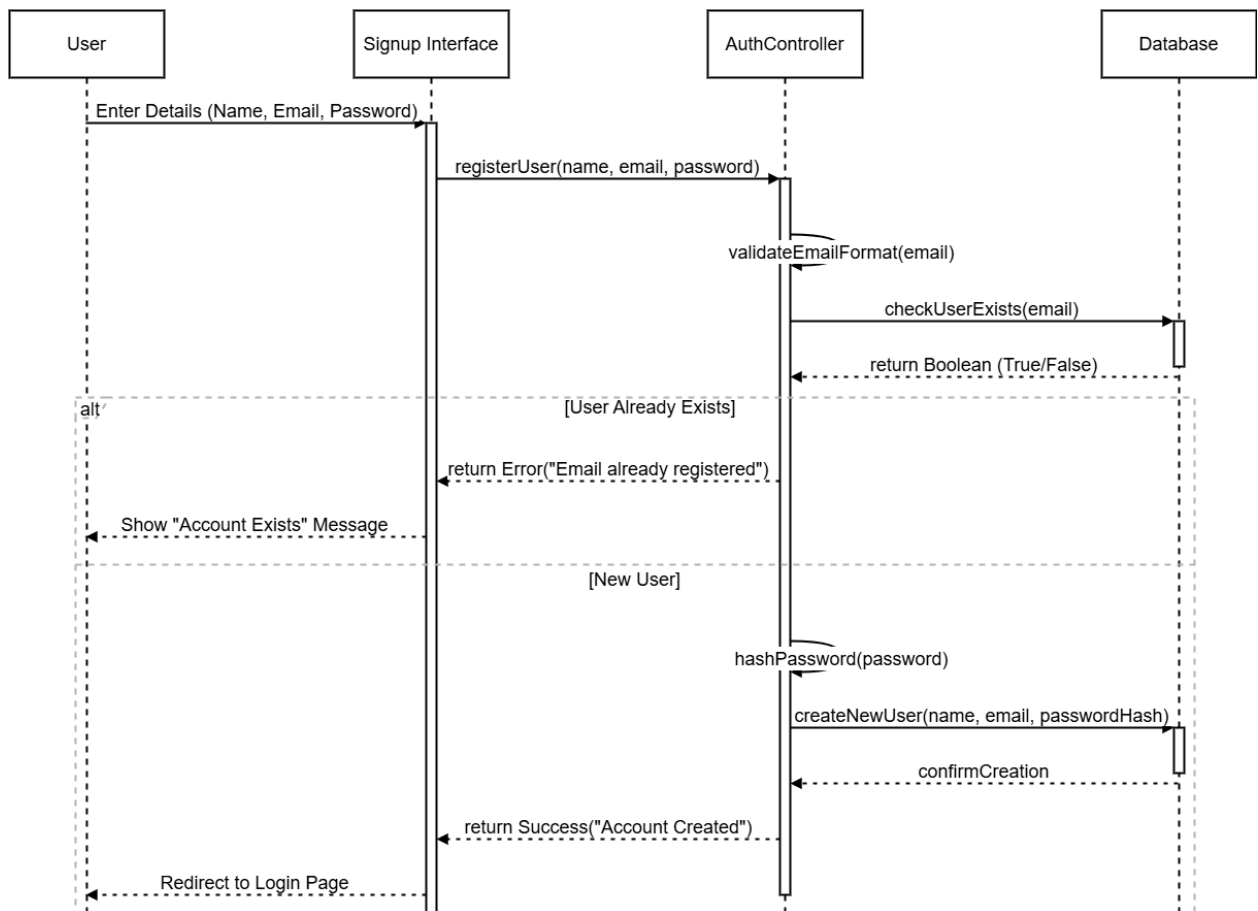
1.Authentication: Login



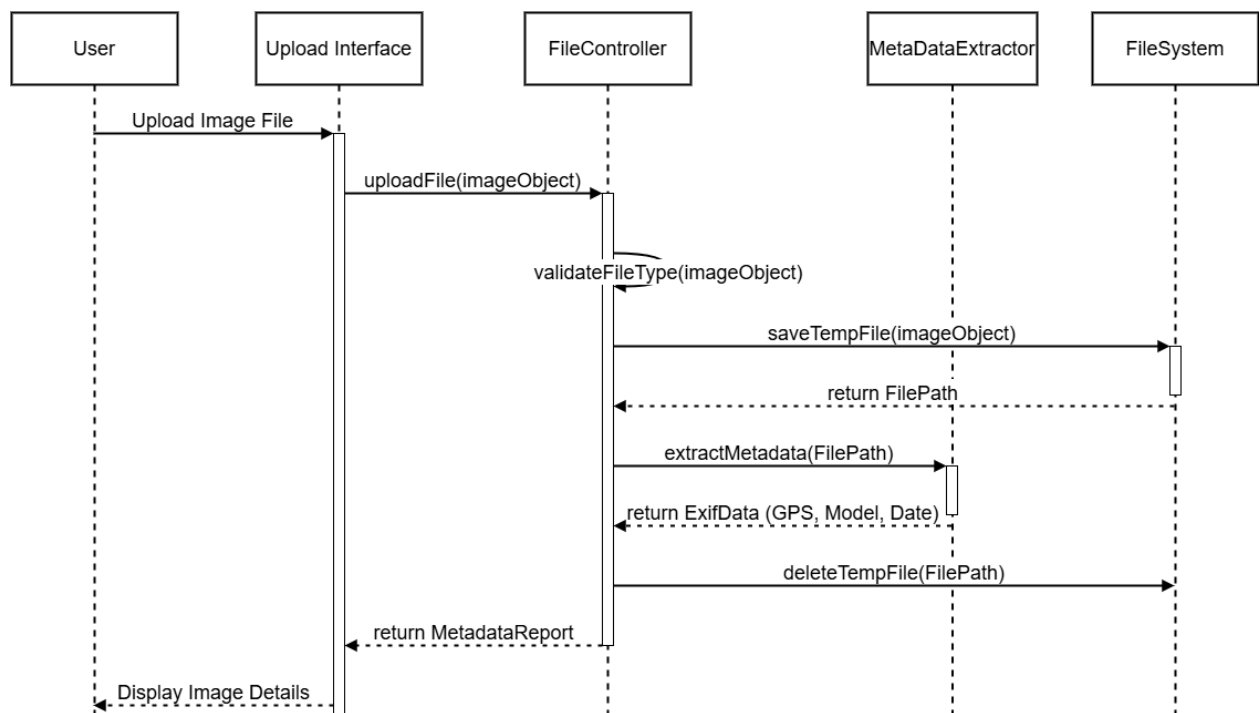
Authentication: Signup



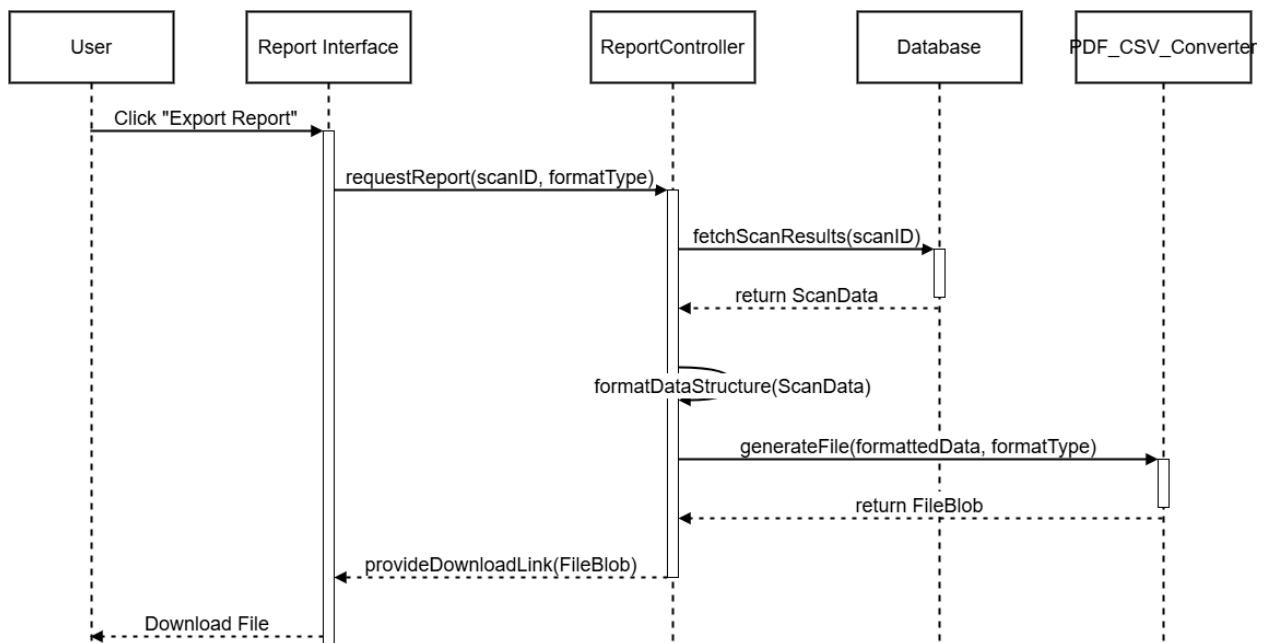
3.OSINT Investigation(Whois,URL)



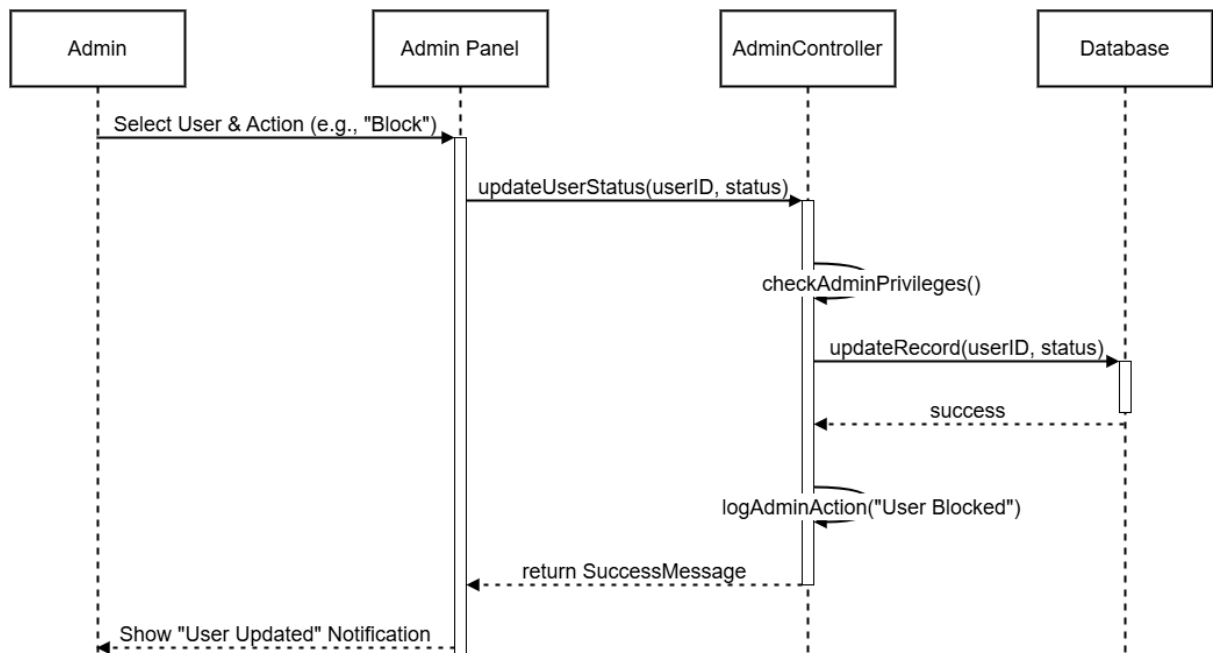
4.OSINT Investigation (Image Analysis)



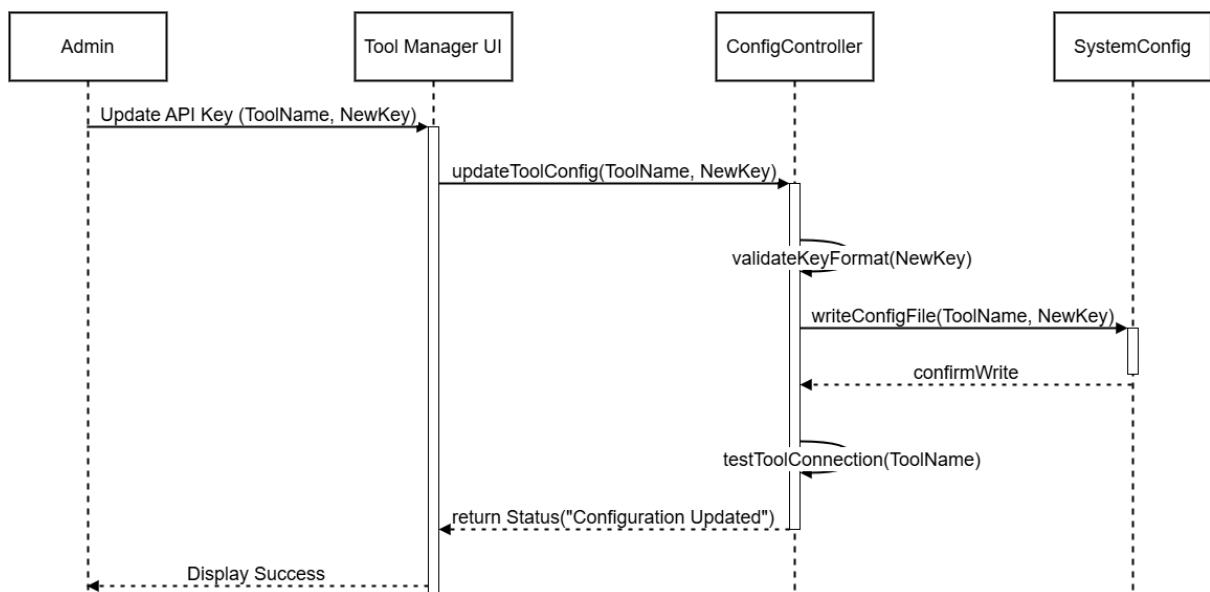
5.Reporting Module



6.Admin: Manage Users



7.Admin: Manage Tools



Entity Relationship Diagram

For

Multi-Vector OSINT Analyzer

Version 1.0

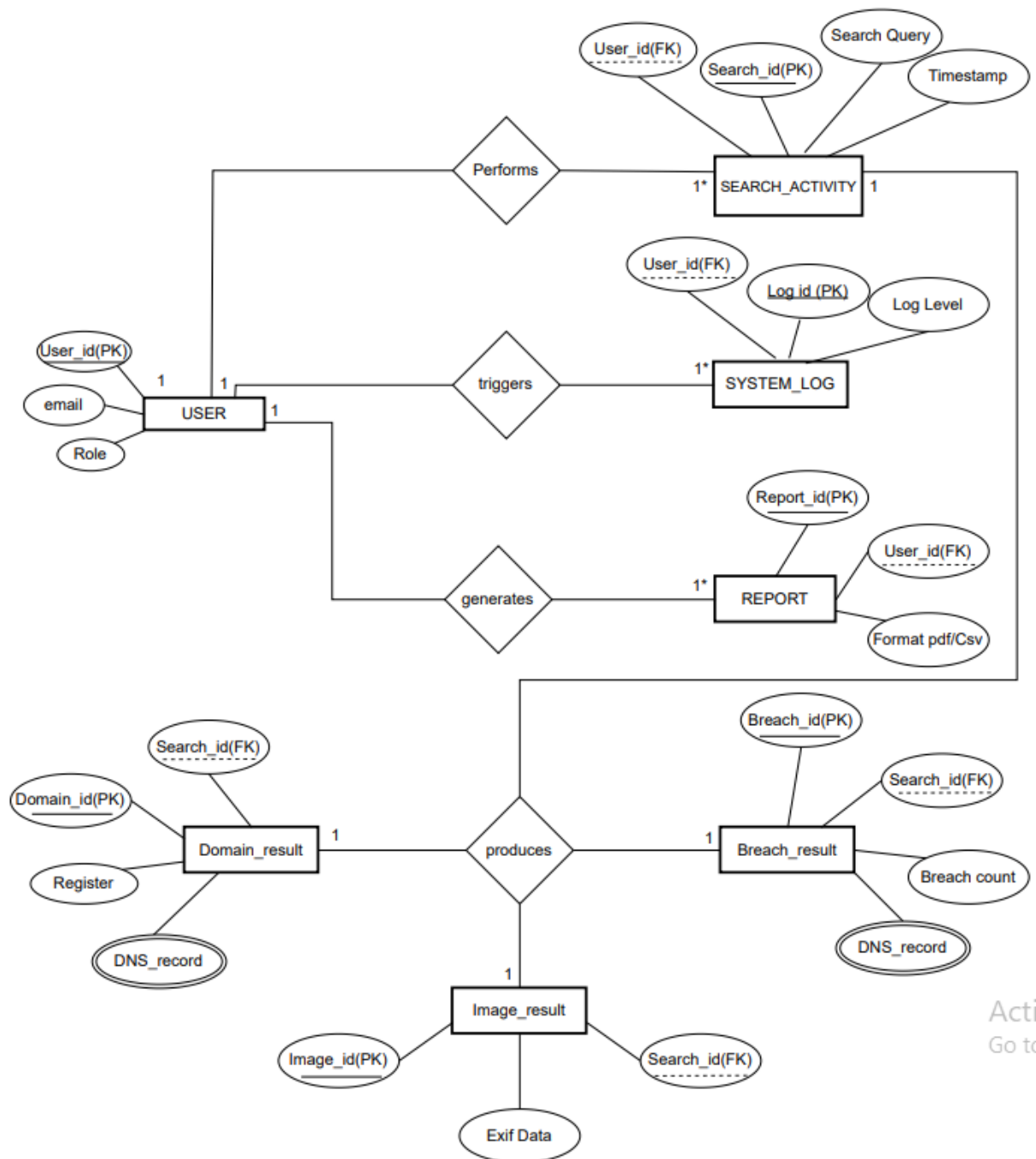
Prepared by

Mansoor Khan

&

Atif Mehmood

11th Jan, 2026



Activate W
Go to Settings

Deployment diagram

Multi-Vector OSINT Analyzer

Version 1.2

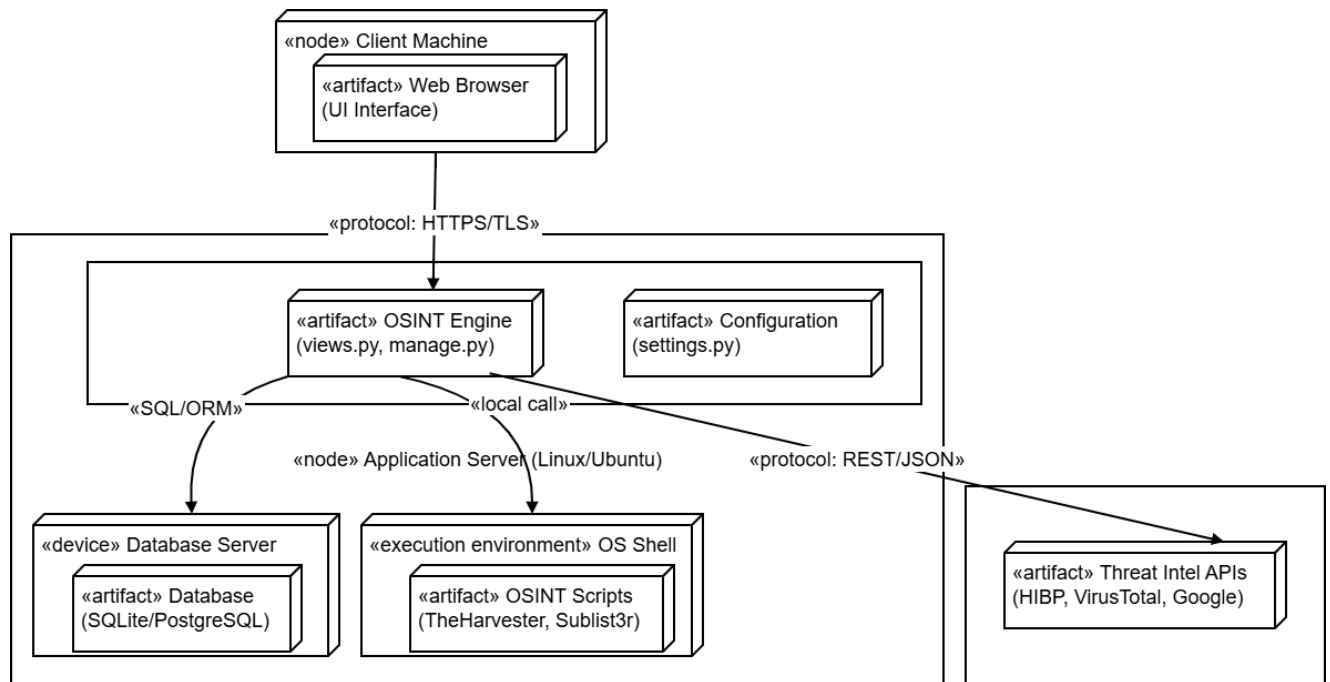
Prepared by

Mansoor Khan

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Atif Mehmood

11th Jan, 2026



Activity Diagram

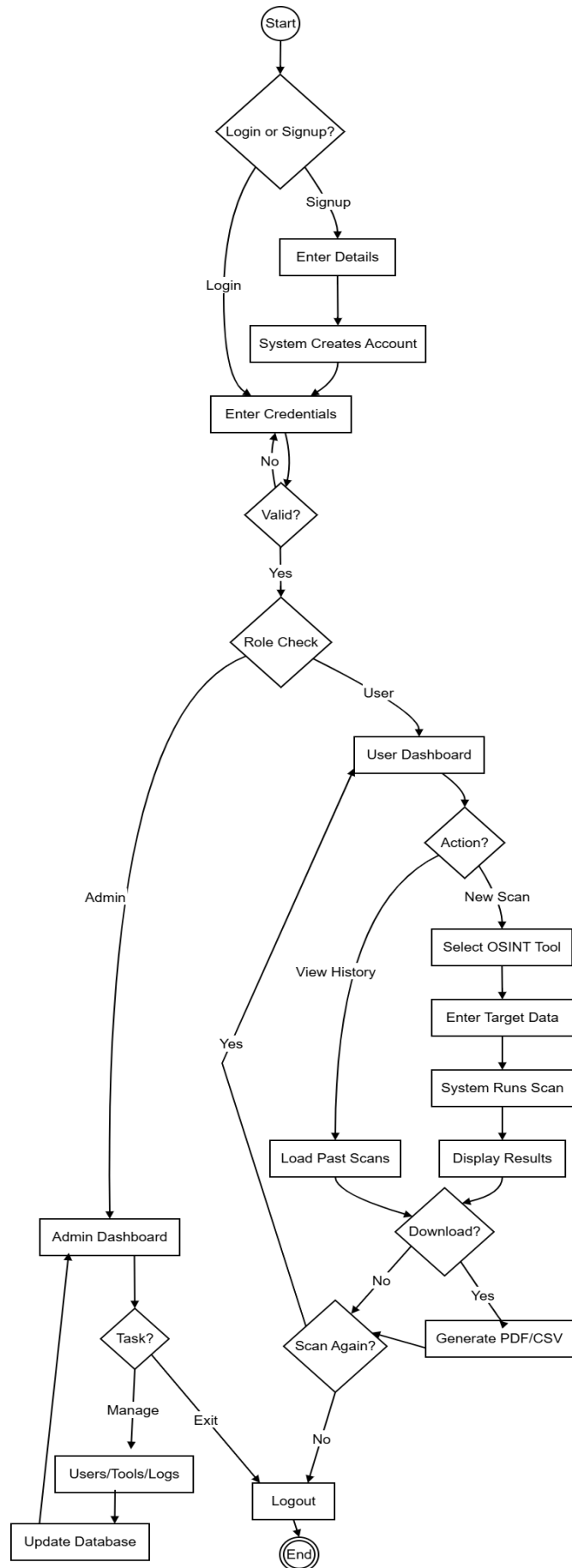
For

Multi-Vector OSINT Analyzer

Version 1.0

Prepared by
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22th Jan, 2026



3.CHAPTER

CONSTRUCTION

Class Diagram

Multi-Vector OSINT Analyzer

Version 1.0

Prepared by

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22th Jan, 2026

